

Why PHP 8.4 and Laravel 13+

A Reproducible Runtime for This Course

Learning Goals

- State the course baseline for PHP, Laravel, and Composer versions.
- Explain why the course requirements are stricter than Laravel 13's official minimum.
- Explain why a shared runtime makes exercises reproducible.

Course Baseline

Every Module 2 Laravel exercise expects:

PHP	8.4.0 or newer
Laravel	13.0.0 or newer
Composer	2.8.0 or newer (major version 2)

These are **course requirements**. They are deliberately stricter than Laravel 13's official minimum PHP version.

Laravel's Official Requirement

Laravel 13 officially supports PHP 8.3 through 8.5 and its Symfony dependencies allow either major line:

```
symfony/console: ^7.4 || ^8.0  
symfony/error-handler: ^7.4 || ^8.0
```

- Symfony 7.4 works on PHP 8.3. Symfony 8 requires PHP 8.4 or newer.

Why PHP 8.3 Is Not Our Baseline

Composer resolves dependencies for the PHP environment where it runs.

- On PHP 8.3, it can select Symfony 7.4.
- On PHP 8.4 or 8.5, it can select Symfony 8.
- `composer.lock` preserves the selected versions.
- A lock file containing Symfony 8 cannot be installed on PHP 8.3.

The framework may support PHP 8.3 while a particular resolved dependency set requires PHP 8.4.

The Failure Students See

```
Your requirements could not be resolved to an installable set of packages.
```

Typical cause:

```
symfony/error-handler 8.x requires php >= 8.4  
your php version (8.3.x) does not satisfy that requirement
```

The issue is the combination of runtime and resolved dependencies, not simply "Laravel 13 always requires PHP 8.4."

Why the Course Uses PHP 8.4+

PHP 8.4 supports Symfony 7.4 and 8, keeps macOS and WSL2 aligned, and avoids direct Symfony constraints or a return to Laravel 12.

Why the Course Uses Laravel 13+

Module 2 is taught against the current Laravel generation rather than the old Laravel 12 skeleton.

- `run1-2.sh` intentionally creates the current application skeleton:

```
composer create-project laravel/laravel student-api
```

- The generated project must report Laravel 13.0.0 or newer before setup continues.